

SimplifyIt: Automatic Simplification of Text

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Through this project we aim to build an automatic text simplification system capable of simplifying medical texts or clinical letters. These texts are usually written for other physicians to understand to communicate the medical needs of a patient with the intention of being as specific as possible. This usually make them very technical and not always easy to understand for the lay patient. This serves as motivation for the development of such a system. However, there is limited parallel complex-simple data available therefore different techniques which work with what data is available were investigated.

Additional Key Words and Phrases: Automatic Text Simplification, Medical Domain

1 INTRODUCTION

1.1 Background

Text simplification is the task of re-writing a sentences to make it easier to understand whilst preserving the semantics of the original. This benefits younger people, people reduces ability to learn or those still not fluent in the language to understand the original text. There seems to be two main approaches in the field namely lexical simplification and syntactic simplification . The former is the task of replacing terminology or jargon with simpler words to make it easier to comprehend the meaning [De Belder and Moens 2012]. A step above is performing this on sentence level rather than on a word-by-word basis which tends to give a more readable result but usually simplifies the output to a lesser degree [Zhang et al. 2017]. Another approach is syntactic simplification which looks at the grammar of the text and tries to break it down into easier sentences whilst retaining the meaning of the original. In this project the focus will be mostly on lexical and sentence-level simplification.

1.2 Motivation for this project

In hospitals, patients get clinical reports on tests and examinations. Such texts are written for clinical staff to efficiently and professionally communicate their results in appropriate medical terminology to other professionals, but unfortunately due to the high technicality of the language used they are often not comprehensible by the patients themselves.

There has been effort worldwide to allow citizens to view their results with the introduction of online portals such as *myHealth* in Malta and the *NHS* app in the UK. However this does not mean that medical records are accessible for all, since not everyone can understand them. In fact, A survey on health literacy in Malta ¹ shows that 45.8% of the Maltese population has limited understanding of clinical texts.

Studies [Ross and Lin 2003] [Esch et al. 2016] concluded that at the very least, having the patient understand their records enhances physician-patient communication, which could make the already unpleasant experience of going to a doctor somewhat more less

¹<https://deputyprimeminister.gov.mt/en/CommMentalHealth/Pages/health-literacy-survey.aspx>

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daunting. This has served as motivation for this project, in which ways to simplify such records would be investigated.

1.3 Aims and Objectives

In this final year project, we aim to have an automatic text simplification system capable of simplifying jargon-intensive medical letters into text which is understandable by the lay patient. This can be divided in the following steps:

- (1) Collecting parallel complex-simple medical text.
- (2) Researching different text simplification systems and identifying which machine learning model would perform best in the medical domain.
- (3) Developing the system and training the model on the data found
- (4) A similar model which employs a phrase-table is to be developed for comparison.
- (5) Evaluating our system by comparing the phrase-table model with the baseline using an objective evaluation metric. A comparison will also be brought out to what has been previously implemented.

We hope that following these goals we develop a system which can give accurate simplifications with minimal semantic loss.

2 LITERATURE REVIEW

2.1 Text Simplification approaches on the General Domain

Most machine text simplification (MTS) approaches in literature consider the task as a monolingual machine translation (MT), which converts the original text into one which is more comprehensible. Both statistical (SMT) and neural machine translation (NMT) techniques have been tried and achieved satisfactory results. Deep neural models and more recently transformers have been looked into to perform the task and were found to give promising results.

Traditionally, rule-based approaches have been taken to perform simplification [Vickrey and Koller 2008]. However, this requires the laborious process of defining high-level rules which has to be done by a human, which led for others to look into data-driven techniques. Several different improvements and adaptations have been tried on the above. [Zhu et al. 2010] have used a tree-based model to perform operations "splitting", "dropping", "re-ordering" and "substitution" which used together output a simpler version of the original text. [Coster and Kauchak 2011] have employed a phrase-based system for monolingual SMT for simplification. Such a model simplifies translates sequences of words at a time rather than going word by word. This means that the model will also take into account the order in which words occur.

More recently, as in fields of AI and natural language processing, neural MT techniques based on sequence-to-sequence models have gained popularity and have shown to give significantly good results. Word-level NMT based simplification models are sufficient to simplify individual words but can produce a sentence which is not syntactically correct due to differences in the meaning of the simpler

synonyms. Conversely, simplification models at sentence-level often outputs an unchanged sentence. [Zhang et al. 2017] proposed performing simplification at two steps which performs simplification both at word-level as well as a full sentence. This has the advantage of giving better lexical simplifications which follow correct grammar. The two steps performed are firstly to identify complex words are replace them with simpler terms (Lexical simplification). Then a constrained seq2seq neural sentence generation model is used to produce an already simple sentence from the identified words. For this model they used a dataset of 123,626 parallel complex-simple sentences.

2.2 Different Approaches for the Medical Domain

Early work on simplifying medical records mostly focus on lexical simplification of the text [Kandula et al. 2010] [Qenam et al. 2017]. This is usually done by finding complex (complex here means that it contains medical terminology) words, phrases or sentences, and using an external phrase table or thesaurus to find simpler semantically equivalent replacements. A popular metathesaurus used for this task is the Unified Medical Language System (UMLS). Most also give importance to syntactic simplification, in which longer sentences are split into two or more shorter ones which are usually more straight to the point.

When it was discovered that the problem of TS can be posed as monolingual MT, more data-driven approaches were looked into for simplification. However, there does not seem to be as much data specific to medical records as one would find in other industries.

To the best of our knowledge there were five main studies on data-driven MTS in the medical domain [Adduru et al. 2018], [Shardlow and Nawaz 2019], [Van den Bercken et al. 2019], [Van et al. 2020], [Pattisapu et al. 2020]. Current medical TS approaches can be summarised to be one of the following or a combination of the below, keeping in consideration the limited resources available:

- (1) Perform lexical simplification
- (2) Make use of a neural TS model trained on generic English - simple English parallel data
- (3) Perform syntactic simplification

[Adduru et al. 2018] was one of the first to look into data-driven approaches for medical TS. The authors created a new dataset, which unfortunately is not publicly available, from aligned complex - simple sentence pairs based on parallel data from webmd, medicinenet, Mayo clinic and English Wikipedia articles aligned with their corresponding ones from simple English Wikipedia. This dataset is made up of 2493 aligned medical sentences. Using this dataset the authors train an encoder-decoder sequence-to-sequence model for simplification. They used the BLEU, METEOR and TER metrics for evaluation. Their simplification model achieved the following scores; 9.9 ± 1.6 BLEU, 10.6 ± 0.8 METEOR, 97.7 ± 2.9 TER.

Due to data limitations, [Shardlow and Nawaz 2019] combine a neural approach with lexical simplification using a phrase table. They used a pre-developed neural TS system [Nisioi et al. 2017] developed on OpenNMT trained on aligned English - Simple English Wikipedia data. The base-line model by [Nisioi et al. 2017] uses two LSTM layers with 500 hidden units and hidden states of size 500. With a probability of dropout of 30%. The authors used data gathered

from i2b2 and MIMIC as a test set for the above-mentioned model, on which they obtained the following scores; FK 6.49, Gunning Fox 12.35 and Coleman Liau 9.04 on the data from i2b2; FK 4.84, Gunning Fox 7.36 and Coleman Liau 5.90 on the data from MIMIC; and only 5.8% of the sentences were simplified to be beneficial. With human analysis of the text it was noted that most simplifications were ones which were not specific to medical terminologies. Therefore due to medical simplification data constraints they decided use a phrase table to simplify specific medical keywords while using the neural model for simplifying the sentence. Rather than greedily substituting all complex words found in the thesaurus, the phrase table was used in conjunction with the neural model so that lexical simplification would only be performed where it would yield a better simplification. By human evaluation, it has been seen that such method gives a better result.

Further studies on the topic where undertaken by [Van den Bercken et al. 2019]. The authors use the NMT-based model for simplification developed by [Nisioi et al. 2017] as a baseline, and compare it to another neural model with a similar architecture to the baseline but is trained on the general simplification dataset developed by [Hwang et al. 2015] combined with a medical dataset collected also from by the authors. This dataset consists of only medical articles on Wikipedia and sentences containing at least 1 medical word were considered part of the dataset. When compared to the one presented by [Adduru et al. 2018], it is somewhat larger in quantity containing 3,797 complex-simple sentence pairs, however they are both very small. Unfortunately there does not seem to be any information on how the two datasets compare in quality. They evaluated their models using the BLEU and SARI automated metrics but also human evaluation. Their model obtained an average score of 45.93 BLEU and 33.10 SARI.

[Van et al. 2020] investigate the use of pre-trained neural language models (PTLM) to create an autocomplete model for MTS rather than a fully automatic TS system which gives full control to the user on the next word, arguing that it makes sense in such a sensitive domain. They compiled a new corpus of complex-simple medical sentence pairs extracted from the general dataset presented by [Kauchak 2013]. To compile the corpus, the considered sentences which fall under a medical title and contain at least 4 medical word, which is stricter than previous datasets. They present an ensemble model which combines BERT, RoBERTa, XLNet and GPT-2. They use the data collected to fine-tune the model. Unfortunately direct comparison against other models is not available for this system since the evaluations metrics extensively used for fully automatic models such BLEU and SARI rely on viewing an entire sentence at one and therefore cannot be used for a prediction model like this one. Instead accuracy@N is used as the following scores where achieved at n=5: RoBERTa 73.2, BERT 59.4, XLNet 53.5, GPT-2 56.12

[Pattisapu et al. 2020] try to overcome the setback of limited data available found by [Shardlow and Nawaz 2019] and [Van den Bercken et al. 2019] by proposing a new model which does not require a complex-simple parallel dataset. They instead train a denoising autoencoder, which is a transformer-based encoder-decoder, on text from medical social media blogs, exploiting the fact that most such text is already simple in its nature. Noise is intentionally added to these sentences. The noise serves various purposes but

most importantly it forces the model to use proper grammar in the outputted sentence which results in a more readable result. This is whilst creating a mapping between the original and simple terminology and avoiding semantic loss from simplification. The authors compare their transformer based model with models similar to those developed by [Shardlow and Nawaz 2019] and [Van den Bercken et al. 2019]. They used a test set similar to that used by [Van den Bercken et al. 2019] for evaluation. The denoising algorithm performed the best on almost all metrics obtaining the following scores: 33.60 BLEU, 41.15 SARI, 65.15 ROUGE, 34.20 METEOR. The authors defend their model getting a lower BLEU score than the neural models by arguing that the dataset used for testing contains many sentences which are not changed much and therefore the BLEU metric rewards not changing much in a sentence.

2.3 Dataset

As has been mentioned above various parallel datasets exist for the purpose of text simplification on the general domain, most of which are compiled from English and simple English Wikipedia articles. It can be observed that mainly two general purpose simplification corpora have become the gold standard namely PWKP presented by [Zhu et al. 2010] and the Newsela corpus which was released by [Xu et al. 2015] with the aim of solving various shortcomings of the former; in particular containing considerable inaccurate simplifications and alignment.

However, these have limited data specific to medical terminology and discourse. In response, as discussed above, various datasets have been proposed which although they do not compare with the general corpora in size, they contain only medical sentences. What is considered as a 'medical' sentence varies between datasets. [Van den Bercken et al. 2019] propose a medical dataset which is a subset of an existing corpus on the general domain containing just the medical sentences.

[Adduru et al. 2018] proposed a dataset which uses Wikipedia sentences as a starting point but due to the limited simplification data on the medical sector, they scraped other clinical sites for more data to increase quantity. They name medicinenet and webmd amongst others as web sources for this data. [Devaraj et al. 2021] compiled their corpus by scraping the Cochrane Database of Systematic Reviews, aligning the technical abstract with the PLS (plain language summary). [Shardlow and Nawaz 2019] used existing medical datasets from MIMIC and the i2b2 shared task and human annotation for alignment. Otherwise they are a very small corpus made up of manually aligned or manually simplified sentences created with the intention for them to be used for testing.

2.4 Evaluation

Using SARI and BLEU metrics, a comparison will be brought out to see to what degree, if at all, the model is improved in terms of simplicity by the implementation of the phrase table. For comparison with the solution proposed by [Shardlow and Nawaz 2019], the Flesch-Kincaid and Gunning Fox metrics will be used to compare the output readability.

The BLEU metric is a widely-used metric for machine translation presented by Papineni et al. 2002 which measures adequacy, fluency

and the fidelity of the translated text. When MT models are applied for simplification, it was only natural to apply BLEU for evaluation. The metric compares the n-gram of the source sentence to that of the resultant one and compares the number of co-occurrences by computing the modified n-gram precision. This measurement is used instead of conventional precision as the latter often gives an unrealistically high result and is found by:

- (1) Finding the maximum number of occurrences of a particular word in a translation (count).
- (2) For every reference sentence the number of occurrences of a candidate n-gram is calculated.
- (3) The maximum number of n-gram occurrences is found (max ref count).
- (4) The minimum of count and max ref count is found:
 $Count_{clip} = \min(Count, MaxRefCount)$
- (5) $\sum Count_{clip}(n - gram)$ is computed for all clips
- (6) The modified precision score P_n is found as below

$$\frac{\sum_{C \in Candidates} \sum_{n-gram \in C} Count_{clip}(n - gram)}{\sum_{C' \in Candidates} \sum_{n-gram' \in C'} Count_{clip}(n - gram)}$$

The modified precision score accounts for the adequacy and fluency of the translation. Words which are the same as the references make better adequacy and longer n-gram matches mean a more fluent translation. The Brevity Penalty (BP), an exponential decay, is calculated to account for shorter translations.

$$Brevity\ Penalty = \begin{cases} 1 & \text{if } c > r \\ e^{1-\frac{r}{c}} & \text{if } c \leq r \end{cases}$$

Using these results the BLEU score can be computed by

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^N w_n \log_2 P_n\right)$$

This score ranges from 0 to 100 and a high score means the simplification closely resembles that performed by a human.

Xu et al. 2016 propose a new metric, SARI, specific for text simplification part and has since then become the standard to evaluate such task. They claimed that previously used metrics such as BLEU is not able to judge lexical simplification well enough. SARI stands for "System output against references and against the input sentence". For this metric n-gram precision and recall, $p(n)$ and $r(n)$ respectively are defined as follows:

$$p_{add}(n) = \frac{\sum_{g \in O} \min(\#_g(O \cap \bar{I}), \#_g(R))}{\sum_{g \in O} \#_g(O \cap \bar{I})}$$

$$r_{add}(n) = \frac{\sum_{g \in O} \min(\#_g(O \cap \bar{I}), \#_g(R))}{\sum_{g \in O} \#_g(R \cap \bar{I})}$$

where $\#_g(\cdot)$ is a binary indicator of occurrence of n-grams g in a given set (and is a fractional indicator in some later formulas) and

$$\#_g O \cap \bar{I} = \max(\#_g(O) - \#_g(I), 0)$$

$$\#_g R \cap \bar{I} = \max(\#_g(R) - \#_g(I), 0)$$

SARI considers words which are already simple enough but can be simplified further and therefore rewards further simplification.

$$p_{keep}(n) = \frac{\sum_{g \in I} \min(\#_g(I \cap O), \#_g(I \cap R'))}{\sum_{g \in I} \#_g(I \cap O)}$$

$$r_{keep}(n) = \frac{\sum_{g \in I} \min(\#_g(I \cap O), \#_g(I \cap R'))}{\sum_{g \in I} \#_g(I \cap R')}$$

where R' is the n -gram counts over R with fractions,

$$\#_g I \cap O = \min(\#_g(I), \#_g(O), 0)$$

$$\#_g I \cap R' = \min(\#_g(I), \#_g(R)/r, 0)$$

Since deleting too much of a sentence makes it less readable than not performing the operation:

$$p_{del}(n) = \frac{\sum_{g \in I} \min(\#_g(I \cap \bar{O}), \#_g(I \cap \bar{R}'))}{\sum_{g \in I} \#_g(I \cap \bar{O})}$$

where

$$\#_g I \cap \bar{O} = \min(\#_g(I) - \#_g(O), 0)$$

$$\#_g I \cap \bar{R}' = \min(\#_g(I) - \#_g(R)/r, 0)$$

Using the above the SARI score can be computed as $SARI = (F1_{add} + F1_{keep} + P_{del})$

3 PROPOSED SOLUTION

All the above-mentioned techniques seem to give acceptable with the resources they have available. However a common problem which can be seen is the lack of specifically clinical complex-simple parallel data. All approaches tried to tackle this problem in their own way, where some tried to use a pre-trained model and only perform fine-tuning and testing on the data. Another approach seen was to train a neural model to employ a phrase-table at situation where the text replacement would give a simpler result. A completely different approach was taken by [Pattisapu et al. 2020] who went around the problem by eliminating the need for such data entirely which performed quite well.

Despite being very interesting and worth looking into, due to time and computational limitations an approach similar to [Pattisapu et al. 2020] was passed over for this project. Instead we decided to focus our efforts on investigating the affect of implementing a phrase-table.

A pre-trained neural model similar to that proposed by Nisioi et al. 2017 will be used and fine-tuned on the medical domain using the dataset presented by [Van et al. 2020]. The latter was chosen since to the best of our knowledge, it is the largest medical text simplification dataset. Like the one proposed by [Van den Bercken et al. 2019], this corpus was compiled from Wikipedia sources, however it is more strict in what is considered a medical sentence. This model will be used as a baseline. Another similar model will be used which also implements a phrase-table.

4 CONCLUSION

Through this project we aim to help address the problem of limited health literacy. We will investigate the effect of using a phrase-table for the task of medical text simplification. A comparison will be brought out between the baseline and the baseline + phrase-table.

REFERENCES

- Viraj Adduru, Sadid A Hasan, Joey Liu, Yuan Ling, Vivek V Datla, Ashequl Qadir, and Oladimeji Farri. 2018. Towards dataset creation and establishing baselines for sentence-level neural clinical paraphrase generation and simplification. In *KHD@IJCAI*.
- Will Coster and David Kauchak. 2011. Learning to Simplify Sentences Using Wikipedia. In *Proceedings of the Workshop on Monolingual Text-To-Text Generation*. Association for Computational Linguistics, Portland, Oregon, 1–9. <https://aclanthology.org/W11-1601>
- Jan De Belder and Marie-Francine Moens. 2012. A Dataset for the Evaluation of Lexical Simplification. In *Computational Linguistics and Intelligent Text Processing*. Alexander Gelbukh (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 426–437.
- Ashwin Devaraj, Iain J Marshall, Byron C Wallace, and Junyi Jessy Li. 2021. Paragraph-level Simplification of Medical Texts. *arXiv preprint arXiv:2104.05767* (2021).
- Tobias Esch, Roanne Mejilla, Melissa Anselmo, Beatrice Podtschaske, Tom Delbanco, and Jan Walker. 2016. Engaging patients through open notes: an evaluation using mixed methods. *BMJ open* 6, 1 (2016), e010034.
- William Hwang, Hannaneh Hajishirzi, Mari Ostendorf, and Wei Wu. 2015. Aligning sentences from standard wikipedia to simple wikipedia. In *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 211–217.
- Sasikiran Kandula, Dorothy Curtis, and Qing Zeng-Treitler. 2010. A semantic and syntactic text simplification tool for health content. In *AMIA annual symposium proceedings*, Vol. 2010. American Medical Informatics Association, 366.
- David Kauchak. 2013. Improving text simplification language modeling using unsimplified text data. In *Proceedings of the 51st annual meeting of the association for computational linguistics (volume 1: Long papers)*. 1537–1546.
- Sergiu Nisioi, Sanja Štajner, Simone Paolo Ponzetto, and Liviu P Dinu. 2017. Exploring neural text simplification models. In *Proceedings of the 55th annual meeting of the association for computational linguistics (volume 2: Short papers)*. 85–91.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*. 311–318.
- Nikhil Pattisapu, Nishant Prabhu, Smriti Bhati, and Vasudeva Varma. 2020. Leveraging Social Media for Medical Text Simplification. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 851–860.
- Basel Qenam, Tae Youn Kim, Mark J Carroll, and Michael Hogarth. 2017. Text simplification using consumer health vocabulary to generate patient-centered radiology reporting: translation and evaluation. *Journal of medical Internet research* 19, 12 (2017), e417.
- Stephen E Ross and Chen-Tan Lin. 2003. The effects of promoting patient access to medical records: a review. *Journal of the American Medical Informatics Association* 10, 2 (2003), 129–138.
- Matthew Shardlow and Raheel Nawaz. 2019. Neural text simplification of clinical letters with a domain specific phrase table. (2019).
- Hoang Van, David Kauchak, and GONDY Leroy. 2020. AutoMeTS: The Autocomplete for Medical Text Simplification. *arXiv preprint arXiv:2010.10573* (2020).
- Laurens Van den Bercken, Robert-Jan Sips, and Christoph Lofi. 2019. Evaluating neural text simplification in the medical domain. In *The World Wide Web Conference*. 3286–3292.
- David Vickrey and Daphne Koller. 2008. Sentence simplification for semantic role labeling. In *Proceedings of ACL-08: HLT*. 344–352.
- Wei Xu, Chris Callison-Burch, and Courtney Napoles. 2015. Problems in current text simplification research: New data can help. *Transactions of the Association for Computational Linguistics* 3 (2015), 283–297.
- Wei Xu, Courtney Napoles, Ellie Pavlick, Quanze Chen, and Chris Callison-Burch. 2016. Optimizing Statistical Machine Translation for Text Simplification. *Transactions of the Association for Computational Linguistics* 4 (2016), 401–415. https://doi.org/10.1162/tacl_a_00107
- Yaoyuan Zhang, Zhenxu Ye, Yansong Feng, Dongyan Zhao, and Rui Yan. 2017. A Constrained Sequence-to-Sequence Neural Model for Sentence Simplification. *CoRR abs/1704.02312* (2017). [arXiv:1704.02312](http://arxiv.org/abs/1704.02312) <http://arxiv.org/abs/1704.02312>
- Zhemmin Zhu, Delphine Bernhard, and Iryna Gurevych. 2010. A Monolingual Tree-based Translation Model for Sentence Simplification. *Coling 2010 - 23rd International Conference on Computational Linguistics, Proceedings of the Conference 2*, 1353–1361.